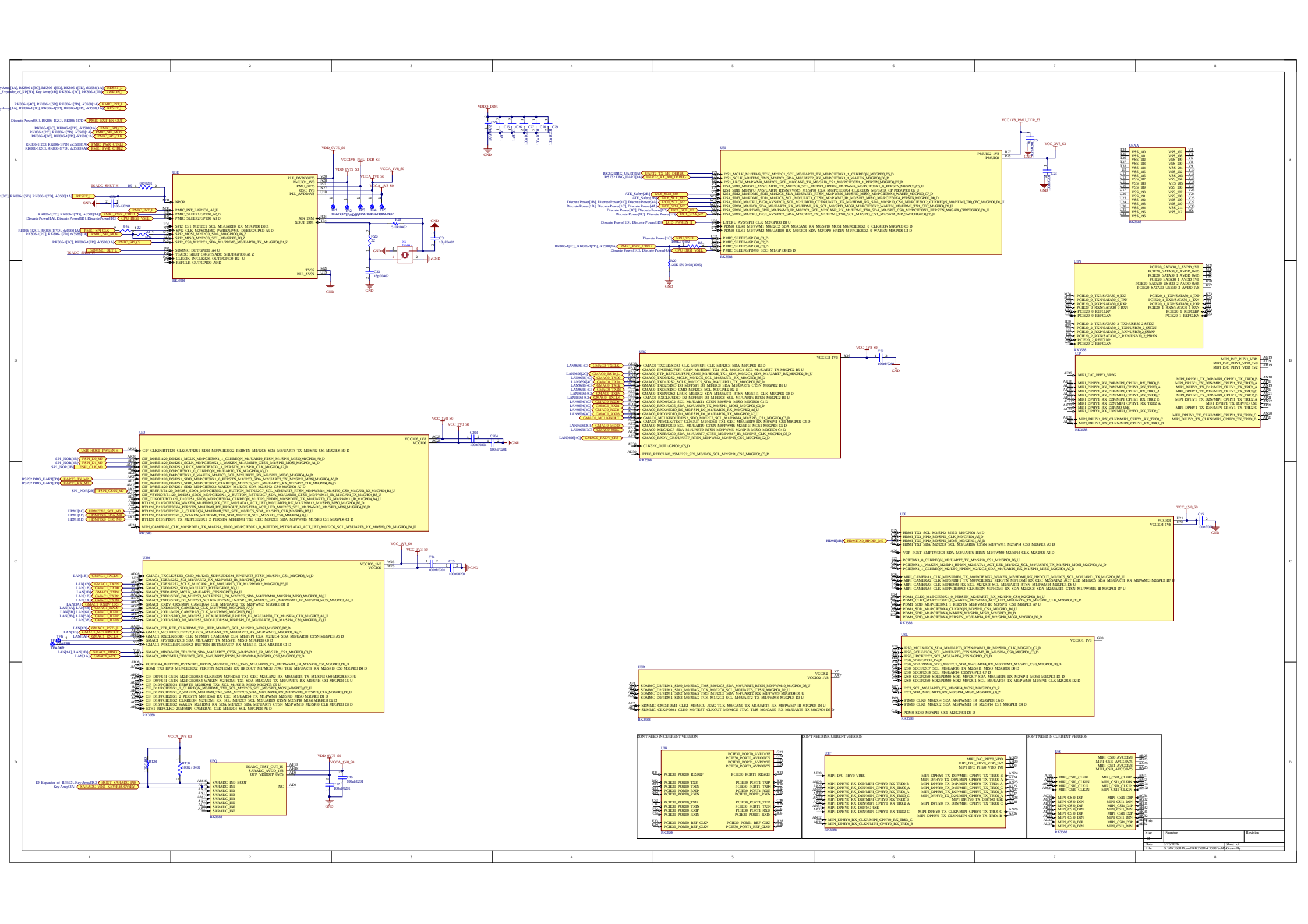
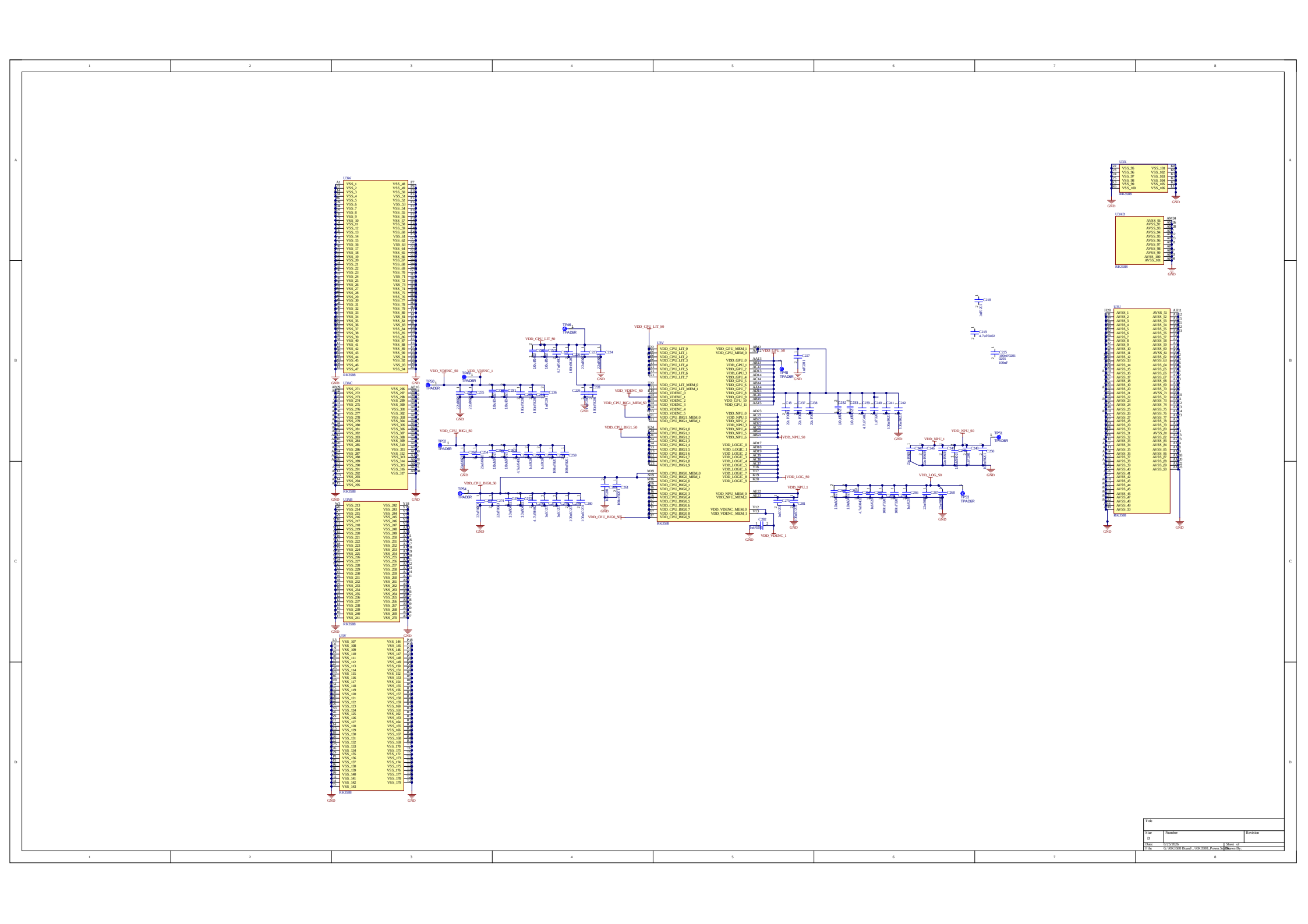
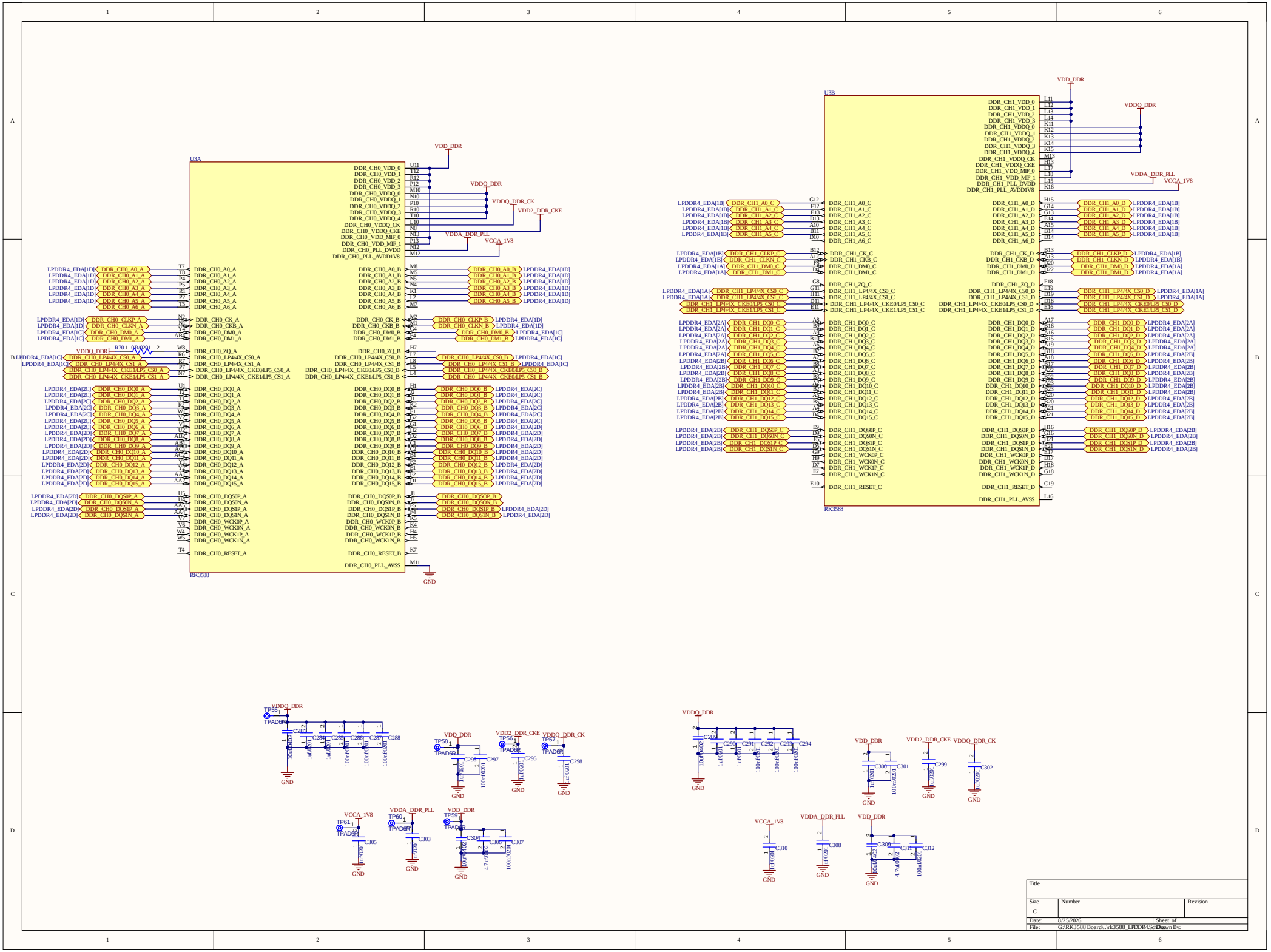
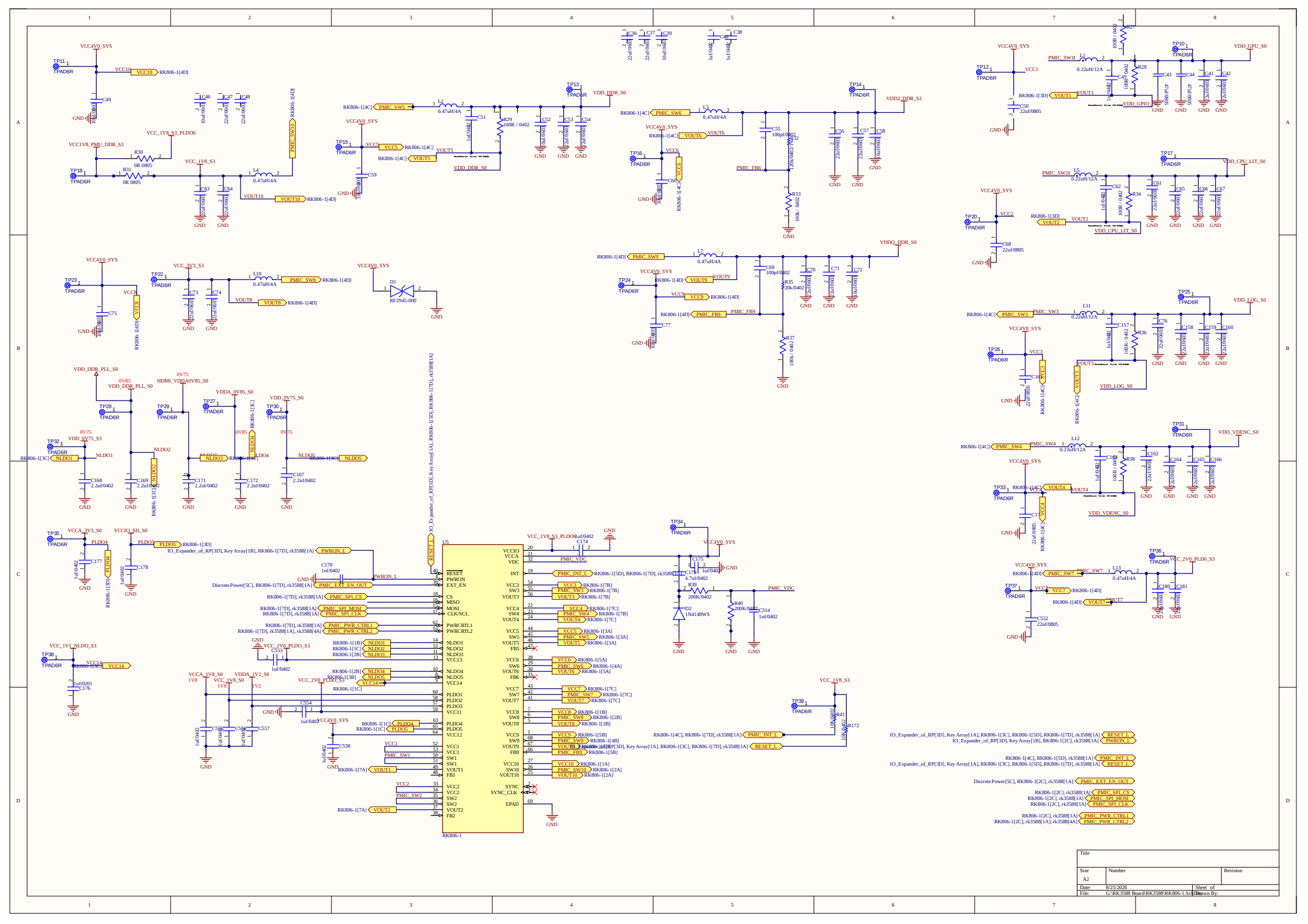
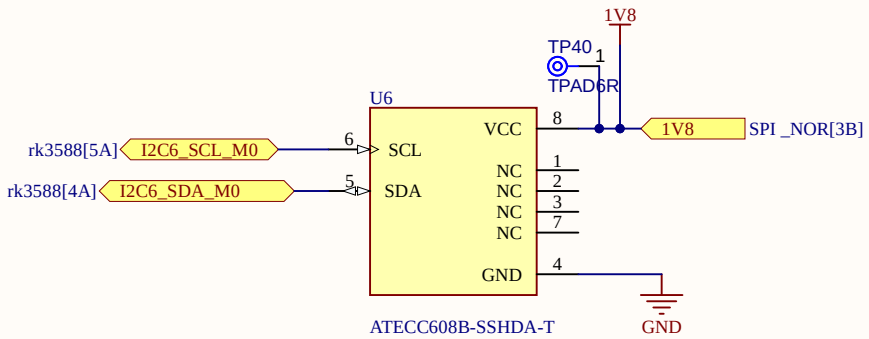

U3E
RK3588
U3U
RK3588
U3M
RK3588
U3Q
RK3588
U3B
RK3588
U3G
RK3588
U3F
RK3588
U3H
RK3588
U3T
RK3588
U3L
RK3588
U3A
RK3588
U3N
RK3588
U3P
RK3588
U3S
RK3588
U3J
RK3588
DON'T NEED IN CURRENT VERSION
DON'T NEED IN CURRENT VERSION
DON'T NEED IN CURRENT VERSION
VDDQ_DDR
VDD_0V75_S0
VCCA_1V8_S0
VCC1V8_PMU_DDR_S3
VCC_3V3_S3
VCC_1V8_S0
GND
PMIC_INT_L
PMIC_SLEEP1
PMIC_SLEEP2
CPU_BIG_VSEL
SPI2_CS1_M2/I2C1_SCL_M1/UART0_RX_M1/GPIO0_B0_Z
SPI2_CLK_M2/SDMMC_PWREN/PMU_DEBUG/GPIO0_A5_D
SPI2_MOSI_M2/I2C0_SDA_M0/GPIO0_A6_Z
SPI2_MISO_M2/I2C0_SCL_M0/GPIO0_B1_Z
SPI2_CS0_M2/I2C1_SDA_M1/PWM5_M0/UART0_TX_M1/GPIO0_B1_Z
SDMMC_DET/GPIO0_A4_U
TSADC_SHUT_OUT/TSADC_SHUT/GPIO0_A1_Z
CLK32K_IN/CLK32K_OUT/GPIO0_B2_U
REFCLK_OUT/GPIO0_A0_D
PMIC_INT_L/GPIO0_A7_U
PMIC_SLEEP1/GPIO0_A2_D
PMIC_SLEEP2/GPIO0_A3_D
TSADC_SHUT_H
RESET_L
CPU_BIG_VSEL
SPI0_CS0_M0
SDMMC_DET_L
PMIC_SPI_CLK
PMIC_SPI_MOSI
PMIC_SPI_MISO
PMIC_SPI_CS
PMIC_INT_L
PMIC_PWR_CTRL1
PMIC_PWR_CTRL2
PMIC_SLEEP1
PMIC_SLEEP2
PMIC_SLEEP3
PMIC_SLEEP4
PMIC_PWR_CTRL3
PMIC_PWR_CTRL4
PMIC_SPI_CS
PMIC_SPI_MISO
PMIC_SPI_MOSI
PMIC_SPI_CLK
PMIC_EXT_EN_OUT
PMIC_SPI_CS
PMIC_SPI_MISO
PMIC_SPI_MOSI
PMIC_SPI_CLK
PMIC_RESET_L
PMIC_PWR_CTRL1
PMIC_PWR_CTRL2
PMIC_PWR_CTRL3
PMIC_PWR_CTRL4
RESET_L
PMIC_INT_L
PMIC_EXT_EN_OUT
PMIC_SPI_CS
PMIC_SPI_MISO
PMIC_SPI_MOSI
PMIC_SPI_CLK
PMIC_RESET_L
PMIC_INT_L
PMIC_PWR_CTRL1
PMIC_PWR_CTRL2
PMIC_PWR_CTRL3
PMIC_PWR_CTRL4
CPU_BIG_VSEL
SPI0_CS0_M0
SDMMC_DET_L
PMIC_SPI_CLK
PMIC_SPI_MOSI
PMIC_SPI_MISO
PMIC_SPI_CS


U39W
U3AC
U3AB
U3Y
U3X
U3AD
U3BJ
U3V
RK3588
VDD_CPU_LIT_S0
VDD_CPU_BIG1_S0
VDD_CPU_BIG0_S0
VDD_VDENC_S0
VDD_VDENC_1
VDD_GPU_S0
VDD_NPU_S0
VDD_NPU_1
VDD_LOG_S0
TP46
TP47
TP48
TP49
TP50
TP51
TP52
TP53
TP54
TP55
TPAD8R
GND
Title
Size
Number
Revision
B
Date: 8/25/2020
Sheet of
File: C:\RK3588 Board\...\RK3588_Power.SchDoc
Drawn By:

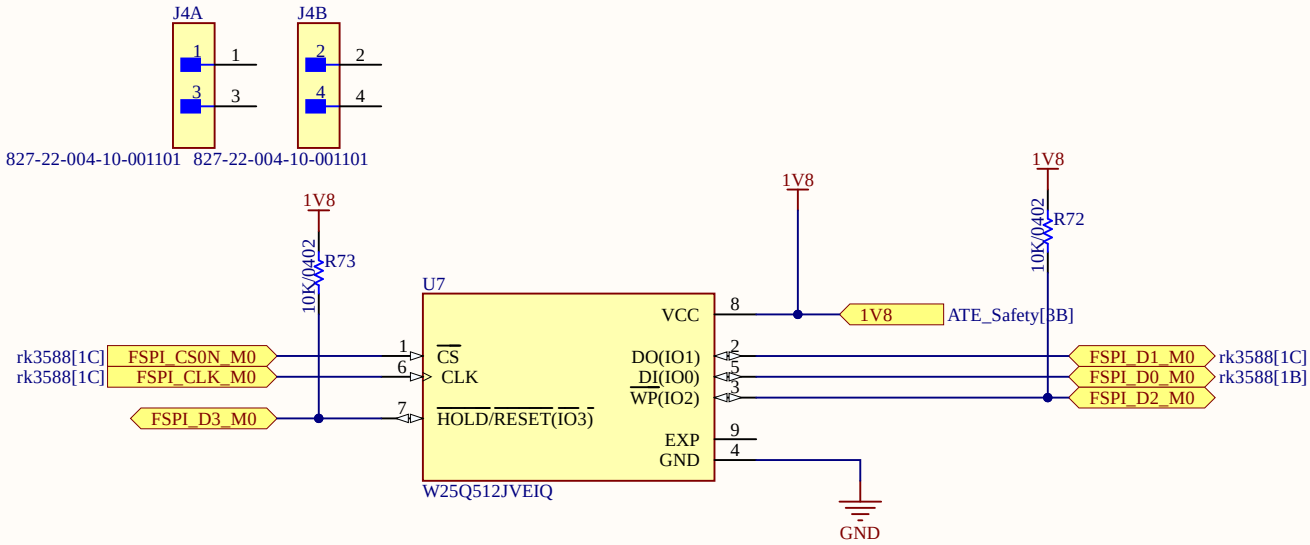

U3A
RK3588
U3B
RK3588
VDD_DDR
VDDQ_DDR
VDDQ_DDR_CK
VDD2_DDR_CKE
VDDA_DDR_PLL
VCCA_1V8
GND
Title
Size
C
Number
Revision
Date: 8/25/2026
Sheet of
File: G:\RK3588 Board\..\rk3588_LPDDR4_SCH_V1.0.sch
Drawn By:



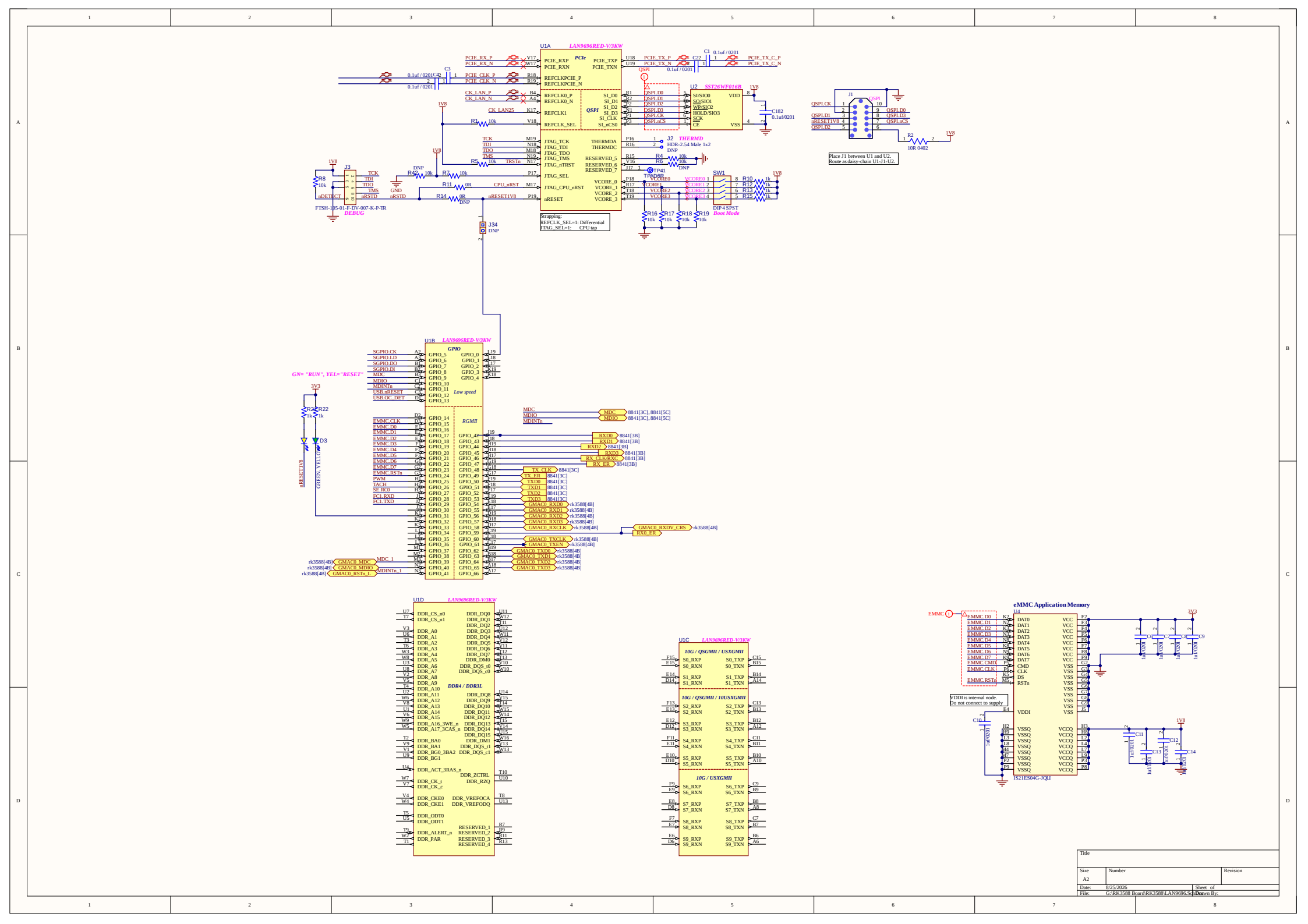
Title	
Size	A2
Number	
Revision	
Date:	8/25/2026
Sheet	of
File:	G:\RK3588 Board\RK3588\RK806-1.SchDoc
Drawn By:	



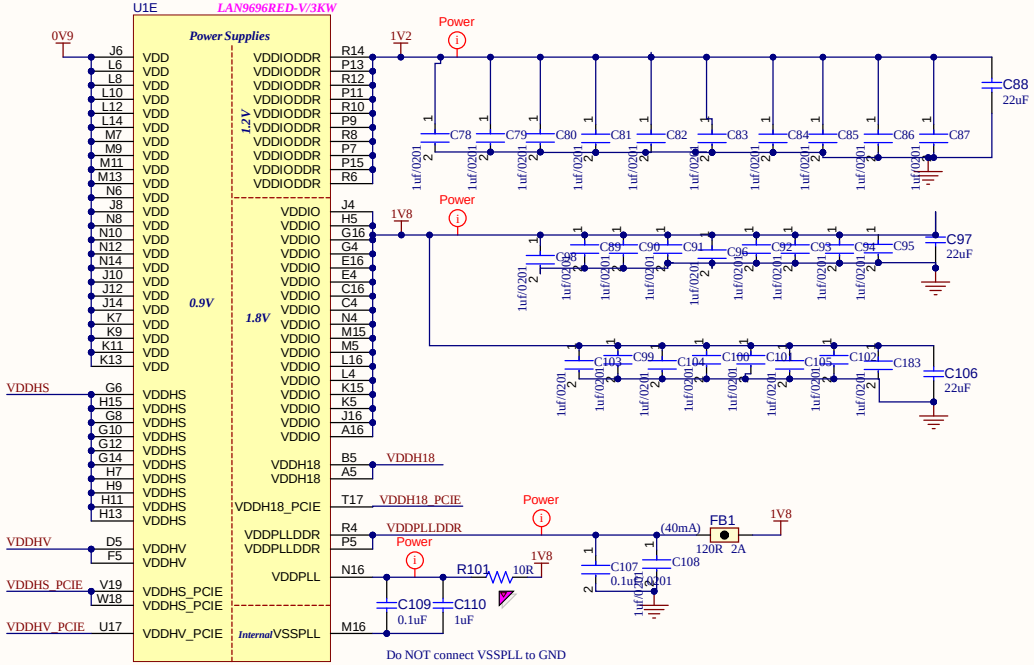
Title			
Size A	Number		Revision
Date:	8/25/2026	Sheet of	
File:	G:\RK3588 Board\..\ATE_Safety.SchDoc	Drawn By:	

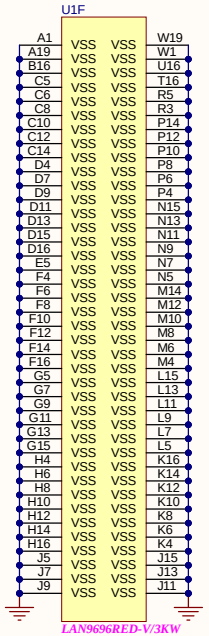


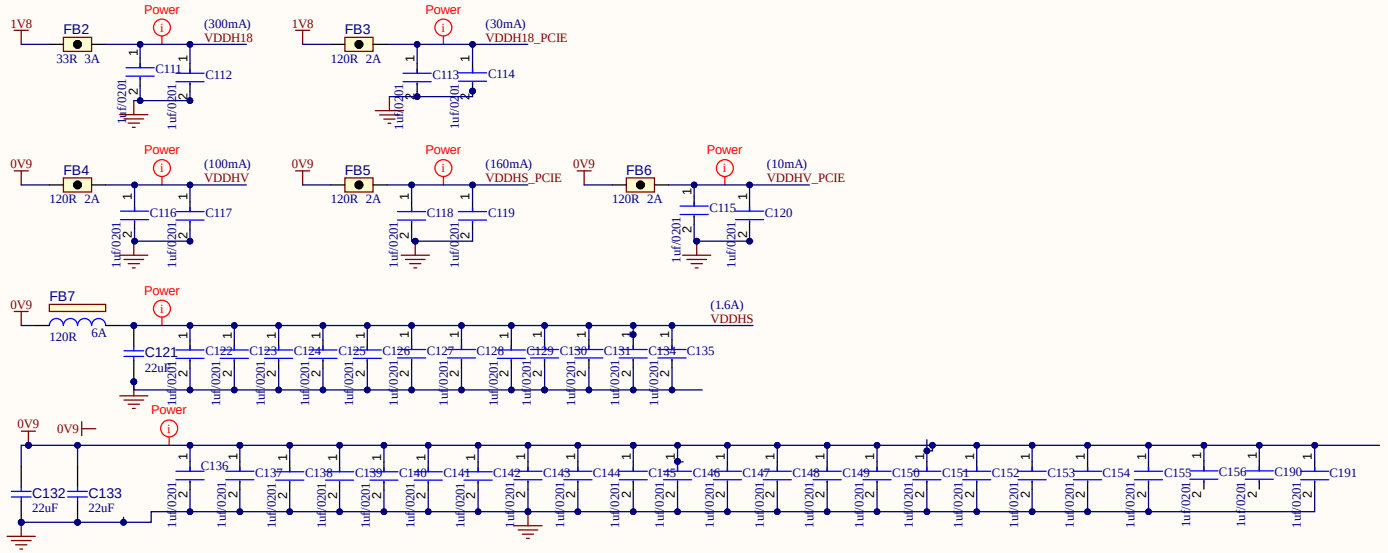
Title		
Size A	Number	Revision
Date: 8/25/2026	Sheet of	
File: G:\RK3588 Board\RK3588\SPI _NOR.SchDoc	Drawn By:	


U1A
LAN9696RED-V/3KW
PCIe
PCIE_RXP
PCIE_RXN
PCIE_TXP
PCIE_TXN
REFCLKPCIE_P
REFCLKPCIE_N
REFCLK0_P
REFCLK0_N
REFCLK1
REFCLK_SEL
QSPI
SI_D0
SI_D1
SI_D2
SI_D3
SI_CLK
SI_nCS0
JTAG_TCK
JTAG_TDI
JTAG_TDO
JTAG_TMS
JTAG_nTRST
JTAG_SEL
JTAG_CPU_nRST
nRESET
THERMDA
THERMDC
RESERVED_5
RESERVED_6
RESERVED_7
VCORE_0
VCORE_1
VCORE_2
VCORE_3
PCIE_RX_P
PCIE_RX_N
PCIE_CLK_P
PCIE_CLK_N
CK_LAN_P
CK_LAN_N
CK_LAN25
TCK
TDI
TDO
TMS
nTRST
TRSTn
CPU_nRST
nRESET1V8
PCIE_TX_P
PCIE_TX_N
PCIE_TX_C_P
PCIE_TX_C_N
QSPI0
QSPI1
QSPI2
QSPI3
QSPI_CK
QSPI_nCS
Strapping:
REFCLK_SEL=1: Differential
JTAG_SEL=1: CPU tap
U2
SST26WF016B
SI/SIO0
SO/SIO1
WP/SIO2
HOLD/SIO3
SCK
CE
VDD
VSS
J1
QSPI
QSPI.CLK
QSPI.D1
nRESET1V8
QSPI.D2
QSPI.D0
QSPI.D3
QSPI.nCS
Place J1 between U1 and U2.
Route as daisy-chain U1-J1-U2.
J2
THERMD
HDR-2.54 Male 1x2
DNP
J3
FTSH-105-01-F-DV-007-K-P-TR
DEBUG
TDI
TDO
TMS
nRSTD
nDETECT
GND
TP41
TP4DBR
SW1
DIP 4 SPST
Boot Mode
VCORE0
VCORE1
VCORE2
VCORE3
J34
DNP
U1B
LAN9696RED-V/3KW
GPIO
Low speed
RGMII
GN= "RUN", YEL="RESET"
nRESET1V8
GREEN, YELLOW
MDC
MDIO
MDINTn
RXD0
RXD1
RXD2
RXD3
RX_CLK/RXC
RX_ER
TX_CLK
TX_ER
TXD0
TXD1
TXD2
TXD3
GMAC0_RXD0
GMAC0_RXD1
GMAC0_RXD2
GMAC0_RXD3
GMAC0_RXCLK
GMAC0_RXDV_CRS
RX0_ER
GMAC0_TXCLK
GMAC0_TXEN
GMAC0_TXD0
GMAC0_TXD1
GMAC0_TXD2
GMAC0_TXD3
GMAC0_MDC
GMAC0_MDIO
GMAC0_RSTn_L
U1D
LAN9696RED-V/3KW
DDR4 / DDR3L
U1C
LAN9696RED-V/3KW
10G / QSGMII / USXGMII
10G / QSGMII / 10USXGMII
10G / USXGMII
eMMC Application Memory
EMMC
U4
IS21ES04G-JQLI
VDDI is internal node.
Do not connect to supply
Title
Size
A2
Number
Revision
Date: 8/25/2026
Sheet of
File: G:\RK3588 Board\RK3588\LAN9696.SchDoc
Drawn By:

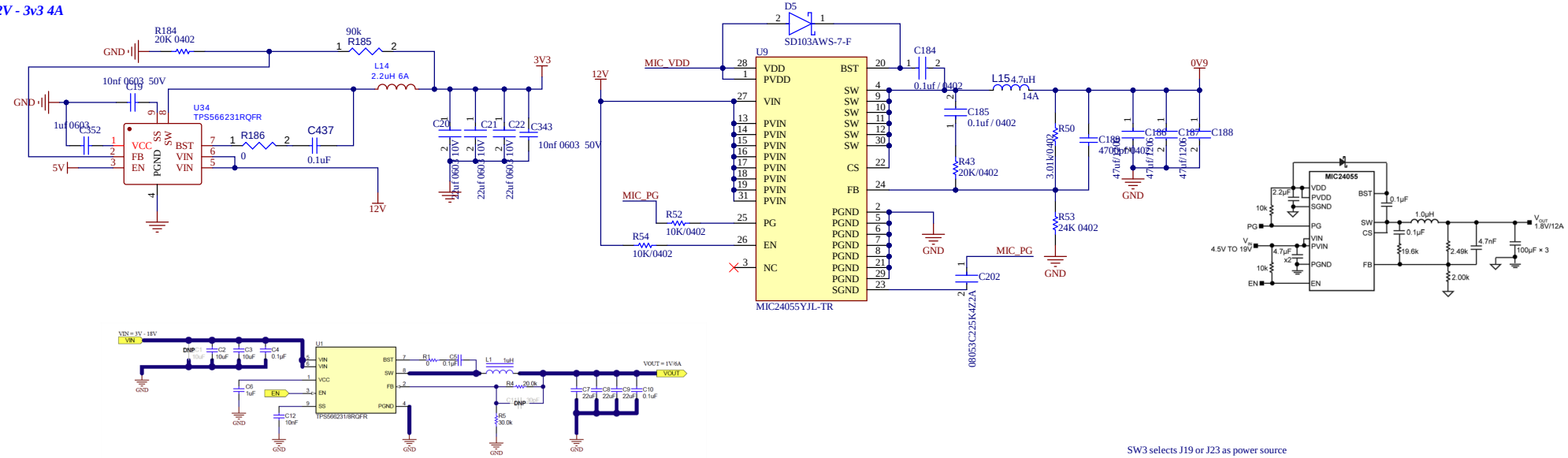
Power supplies:
0.9V 6.7A : LAN9698 VDD(5A)+VDDHS(1.6A)+VDDHV(0.1A)
1.2V 0.5A : LAN9698 VDDIODDR(0.2A)+DDR VDDQ(0.3A)
1.8V 1.7A : LAN9698 VDDH(0.3A)+VDDIO(0.5A)+ZL30732(0.5A)+Other(0.4A)



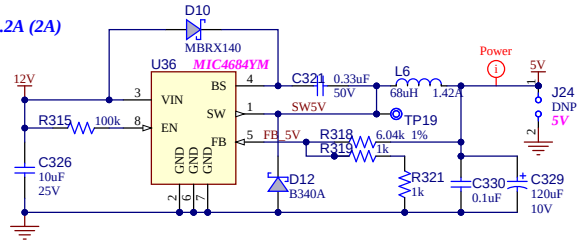




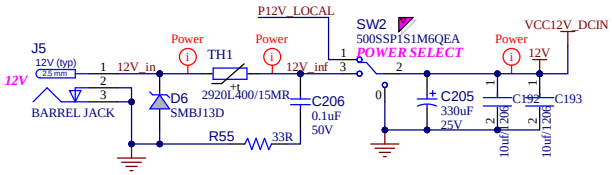
12V - 3v3 4A

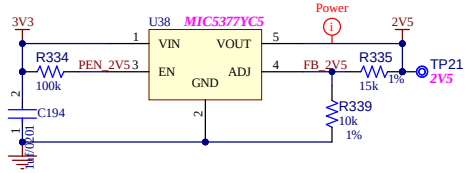


5V @ 1.2A (2A)

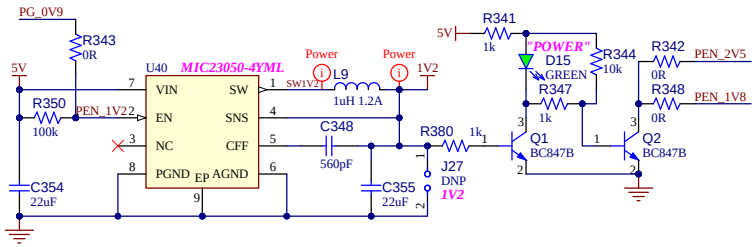


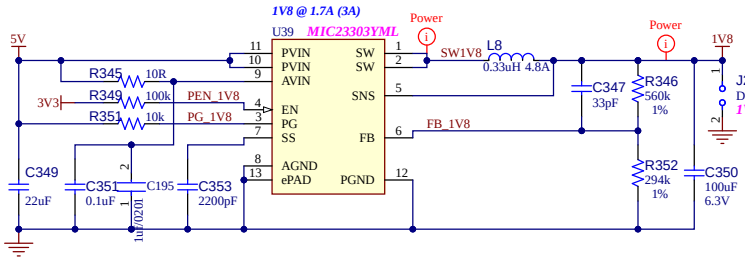
SW3 selects J19 or J23 as power source



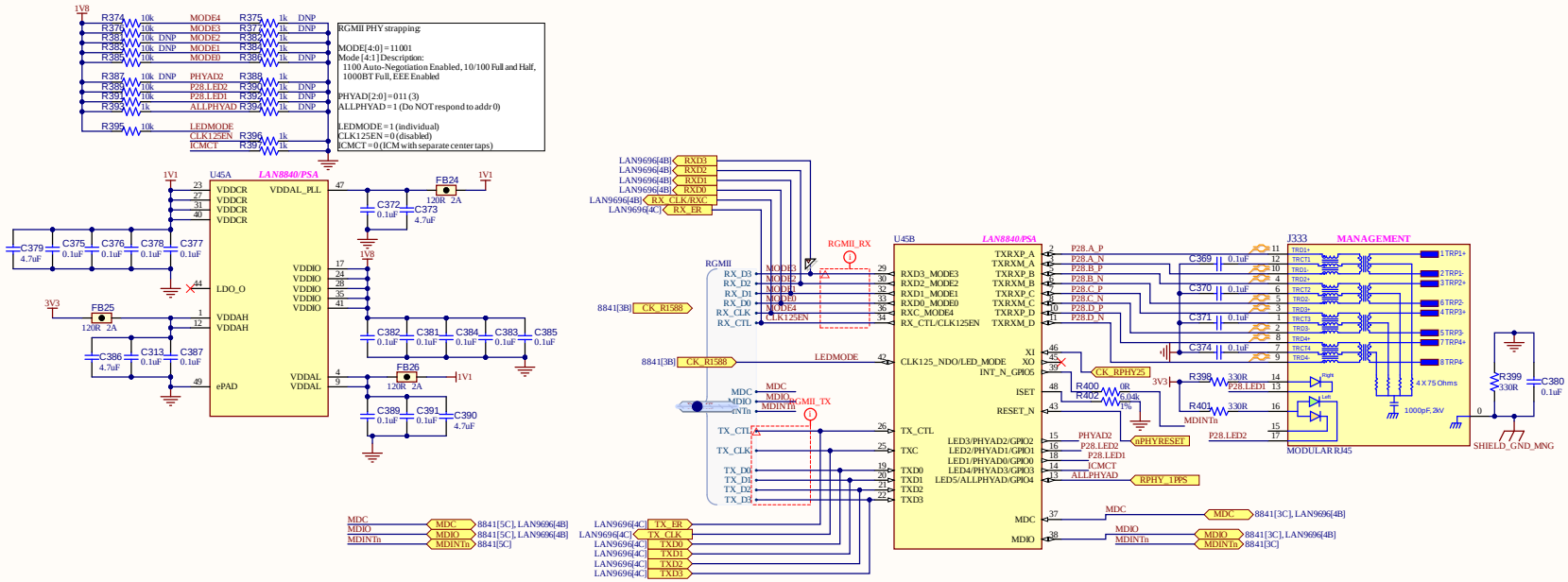


1V2 @ 480mA (600mA)

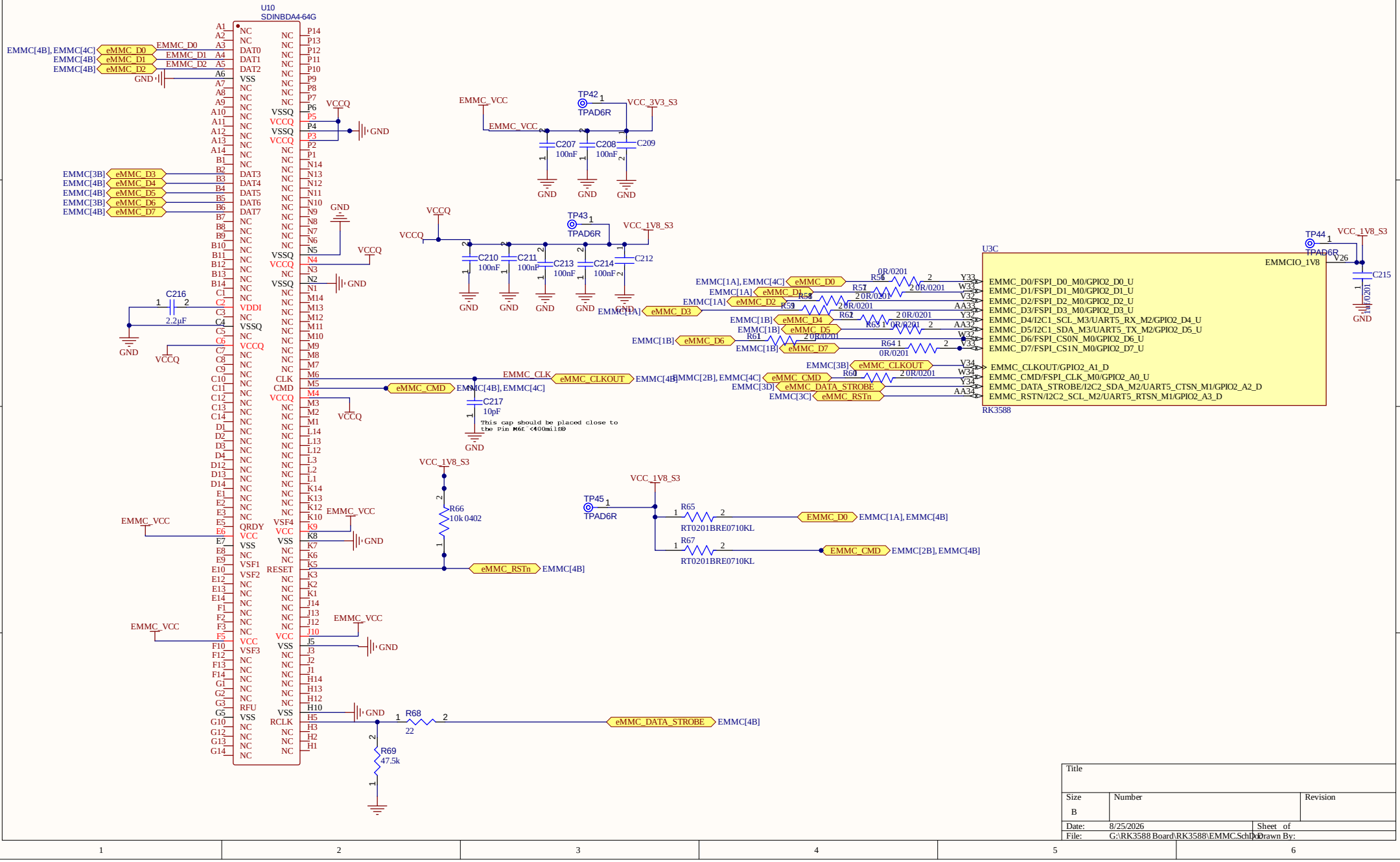


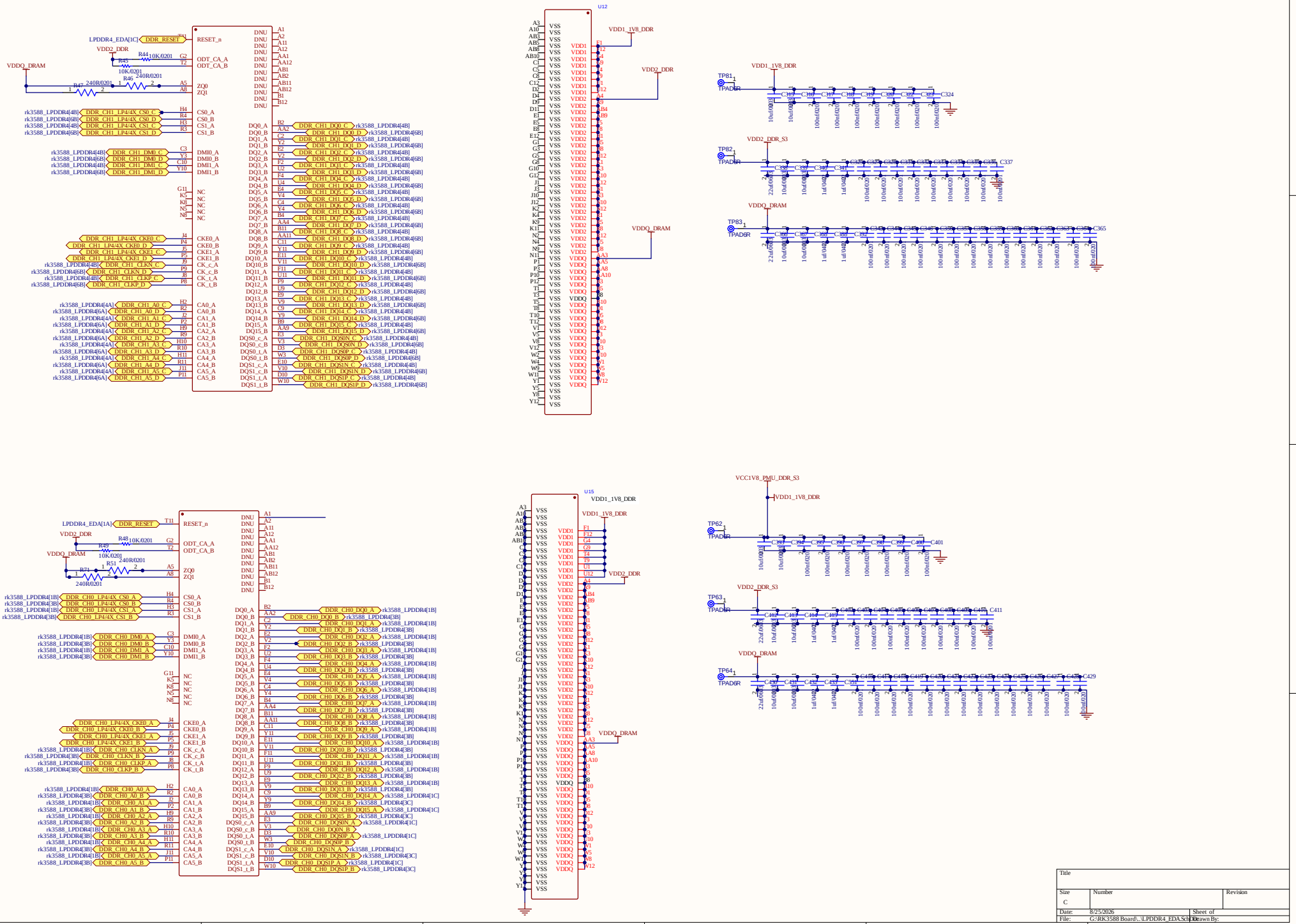


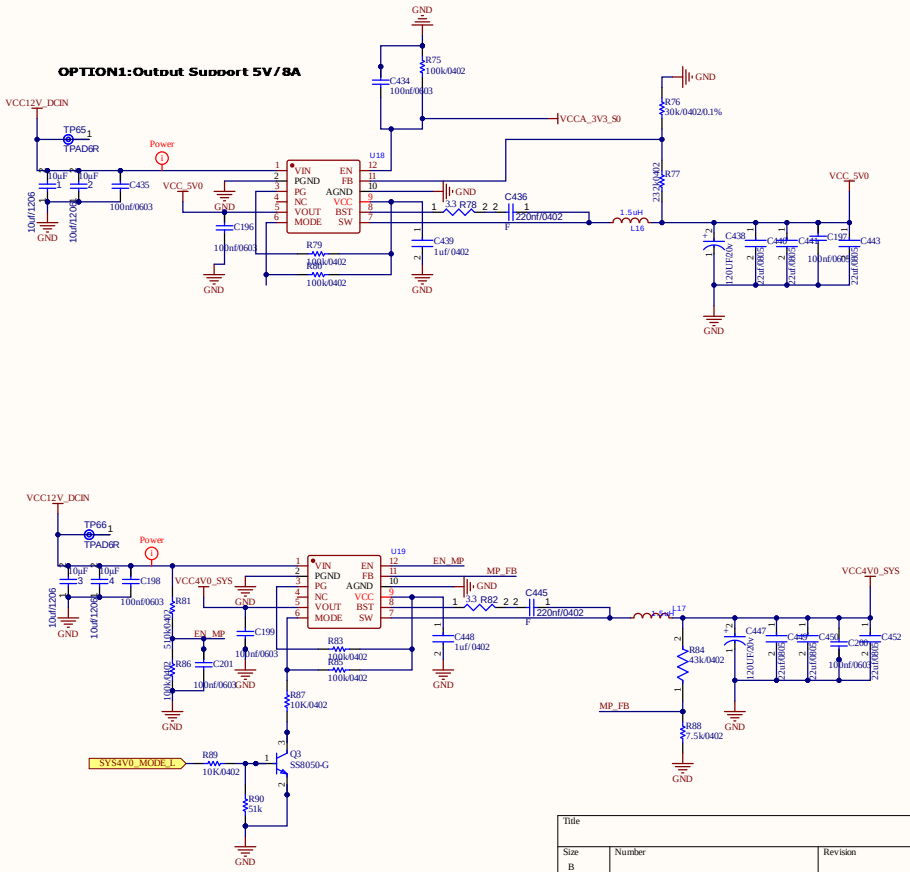
Title		
Size A3	Number	Revision
Date: 8/25/2026	Sheet of	
File: G:\RK3588 Board\..\LAN_Supply.SchDoc	Drawn By:	



Title		
Size: C	Number	Revision
Date: 8/25/2026	Sheet of	
File: G:\RK3588 Board\RK3588\8841.SchDoc	Drawn By:	


U10
SDINBDA4-64G
U3C
RK3588
EMMCIO_1V8
EMMC_D0/FSPI_D0_M0/GPIO2_D0_U
EMMC_D1/FSPI_D1_M0/GPIO2_D1_U
EMMC_D2/FSPI_D2_M0/GPIO2_D2_U
EMMC_D3/FSPI_D3_M0/GPIO2_D3_U
EMMC_D4/I2C1_SCL_M3/UART5_RX_M2/GPIO2_D4_U
EMMC_D5/I2C1_SDA_M3/UART5_TX_M2/GPIO2_D5_U
EMMC_D6/FSPI_CS0N_M0/GPIO2_D6_U
EMMC_D7/FSPI_CS1N_M0/GPIO2_D7_U
EMMC_CLKOUT/GPIO2_A1_D
EMMC_CMD/FSPI_CLK_M0/GPIO2_A0_U
EMMC_DATA_STROBE/I2C2_SDA_M2/UART5_CTSN_M1/GPIO2_A2_D
EMMC_RSTN/I2C2_SCL_M2/UART5_RTSN_M1/GPIO2_A3_D
This cap should be placed close to the Pin M6(<400mil)
Title
Size
B
Number
Revision
Date: 8/25/2026
Sheet of
File: G:\RK3588 Board\RK3588\EMMC.SchDoc
Drawn By:


LPDDR4_EDA[1C]
DDR_RESET
RESET_n
VDD2_DDR
R44 10K/0201
R45 10K/0201
ODT_CA_A
ODT_CA_B
VDDQ_DRAM
R47 240R/0201
R46 240R/0201
ZQ0
ZQ1
rk3588_LPDDR4[4B]
DDR_CH1_LP4/4X_CS0_C
DDR_CH1_LP4/4X_CS0_D
DDR_CH1_LP4/4X_CS1_C
DDR_CH1_LP4/4X_CS1_D
CS0_A
CS0_B
CS1_A
CS1_B
DDR_CH1_DM0_C
DDR_CH1_DM0_D
DDR_CH1_DM1_C
DDR_CH1_DM1_D
DMI0_A
DMI0_B
DMI1_A
DMI1_B
NC
DDR_CH1_LP4/4X_CKE0_C
DDR_CH1_LP4/4X_CKE0_D
DDR_CH1_LP4/4X_CKE1_C
DDR_CH1_LP4/4X_CKE1_D
CKE0_A
CKE0_B
CKE1_A
CKE1_B
DDR_CH1_CLKN_C
DDR_CH1_CLKN_D
DDR_CH1_CLKP_C
DDR_CH1_CLKP_D
CK_c_A
CK_c_B
CK_t_A
CK_t_B
DDR_CH1_A0_C
DDR_CH1_A0_D
DDR_CH1_A1_C
DDR_CH1_A1_D
DDR_CH1_A2_C
DDR_CH1_A2_D
DDR_CH1_A3_C
DDR_CH1_A3_D
DDR_CH1_A4_C
DDR_CH1_A4_D
DDR_CH1_A5_C
DDR_CH1_A5_D
CA0_A
CA0_B
CA1_A
CA1_B
CA2_A
CA2_B
CA3_A
CA3_B
CA4_A
CA4_B
CA5_A
CA5_B
DQ0_A
DQ0_B
DQ1_A
DQ1_B
DQ2_A
DQ2_B
DQ3_A
DQ3_B
DQ4_A
DQ4_B
DQ5_A
DQ5_B
DQ6_A
DQ6_B
DQ7_A
DQ7_B
DQ8_A
DQ8_B
DQ9_A
DQ9_B
DQ10_A
DQ10_B
DQ11_A
DQ11_B
DQ12_A
DQ12_B
DQ13_A
DQ13_B
DQ14_A
DQ14_B
DQ15_A
DQ15_B
DQS0_c_A
DQS0_c_B
DQS0_t_A
DQS0_t_B
DQS1_c_A
DQS1_c_B
DQS1_t_A
DQS1_t_B
DDR_CH1_DQ0_C
DDR_CH1_DQ0_D
DDR_CH1_DQ1_C
DDR_CH1_DQ1_D
DDR_CH1_DQ2_C
DDR_CH1_DQ2_D
DDR_CH1_DQ3_C
DDR_CH1_DQ3_D
DDR_CH1_DQ4_C
DDR_CH1_DQ4_D
DDR_CH1_DQ5_C
DDR_CH1_DQ5_D
DDR_CH1_DQ6_C
DDR_CH1_DQ6_D
DDR_CH1_DQ7_C
DDR_CH1_DQ7_D
DDR_CH1_DQ8_C
DDR_CH1_DQ8_D
DDR_CH1_DQ9_C
DDR_CH1_DQ9_D
DDR_CH1_DQ10_C
DDR_CH1_DQ10_D
DDR_CH1_DQ11_C
DDR_CH1_DQ11_D
DDR_CH1_DQ12_C
DDR_CH1_DQ12_D
DDR_CH1_DQ13_C
DDR_CH1_DQ13_D
DDR_CH1_DQ14_C
DDR_CH1_DQ14_D
DDR_CH1_DQ15_C
DDR_CH1_DQ15_D
DDR_CH1_DQS0N_C
DDR_CH1_DQS0N_D
DDR_CH1_DQS0P_C
DDR_CH1_DQS0P_D
DDR_CH1_DQS1N_C
DDR_CH1_DQS1N_D
DDR_CH1_DQS1P_C
DDR_CH1_DQS1P_D
rk3588_LPDDR4[4B]
rk3588_LPDDR4[6B]
U12
VDD1_1V8_DDR
VDD2_DDR
VDDQ_DRAM
TP81
TP82
TP83
C317
C324
C337
C365
U15
TP62
TP63
TP64
C397
C401
C411
C429
DDR_CH0_RESET
DDR_CH0_LP4/4X_CS0_A
DDR_CH0_LP4/4X_CS0_B
DDR_CH0_LP4/4X_CS1_A
DDR_CH0_LP4/4X_CS1_B
DDR_CH0_DM0_A
DDR_CH0_DM0_B
DDR_CH0_DM1_A
DDR_CH0_DM1_B
DDR_CH0_LP4/4X_CKE0_A
DDR_CH0_LP4/4X_CKE0_B
DDR_CH0_LP4/4X_CKE1_A
DDR_CH0_LP4/4X_CKE1_B
DDR_CH0_CLKN_A
DDR_CH0_CLKN_B
DDR_CH0_CLKP_A
DDR_CH0_CLKP_B
DDR_CH0_A0_A
DDR_CH0_A0_B
DDR_CH0_A1_A
DDR_CH0_A1_B
DDR_CH0_A2_A
DDR_CH0_A2_B
DDR_CH0_A3_A
DDR_CH0_A3_B
DDR_CH0_A4_A
DDR_CH0_A4_B
DDR_CH0_A5_A
DDR_CH0_A5_B
DDR_CH0_DQ0_A
DDR_CH0_DQ0_B
DDR_CH0_DQ1_A
DDR_CH0_DQ1_B
DDR_CH0_DQ2_A
DDR_CH0_DQ2_B
DDR_CH0_DQ3_A
DDR_CH0_DQ3_B
DDR_CH0_DQ4_A
DDR_CH0_DQ4_B
DDR_CH0_DQ5_A
DDR_CH0_DQ5_B
DDR_CH0_DQ6_A
DDR_CH0_DQ6_B
DDR_CH0_DQ7_A
DDR_CH0_DQ7_B
DDR_CH0_DQ8_A
DDR_CH0_DQ8_B
DDR_CH0_DQ9_A
DDR_CH0_DQ9_B
DDR_CH0_DQ10_A
DDR_CH0_DQ10_B
DDR_CH0_DQ11_A
DDR_CH0_DQ11_B
DDR_CH0_DQ12_A
DDR_CH0_DQ12_B
DDR_CH0_DQ13_A
DDR_CH0_DQ13_B
DDR_CH0_DQ14_A
DDR_CH0_DQ14_B
DDR_CH0_DQ15_A
DDR_CH0_DQ15_B
DDR_CH0_DQS0N_A
DDR_CH0_DQS0N_B
DDR_CH0_DQS0P_A
DDR_CH0_DQS0P_B
DDR_CH0_DQS1N_A
DDR_CH0_DQS1N_B
DDR_CH0_DQS1P_A
DDR_CH0_DQS1P_B
rk3588_LPDDR4[1B]
rk3588_LPDDR4[3B]
VCC1V8_PMU_DDR_S3
VDD1_1V8_DDR
VDD2_DDR_S3
VDDQ_DRAM
Title
Size
C
Number
Revision
Date: 8/25/2026
Sheet of
File: G:\RK3588 Board\..\LPDDR4_EDA.SchDoc
Drawn By:



Title		
Size: B	Number:	Revision:
Date: 8/25/2026		Sheet of
File: G:\RK3588 Board\...\BoardSupply.SchDoc		Drawn By:

Note:

TYPEC0_USB20_OTG:

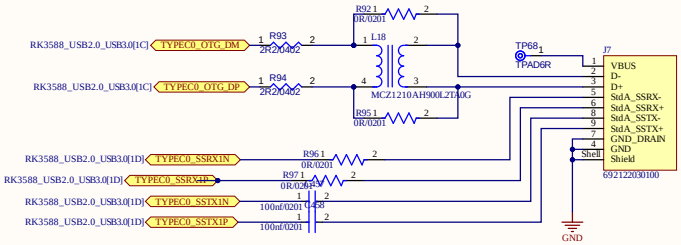
DP/DM:Mask used for download
ID:According to demand,if not used,Leave floating
VBUSDET:Mask provide
REXT:200ohm 1% resistor must be connected externally
Power: Mask supply power

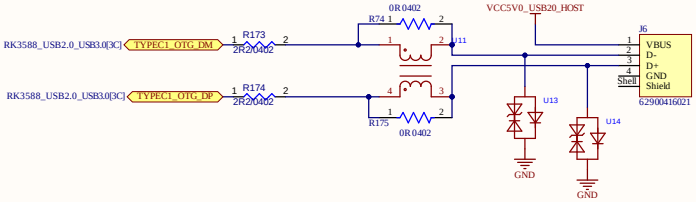
TYPEC1_USB20_OTG: | **USB20_HOST0/USB20_HOST1:**

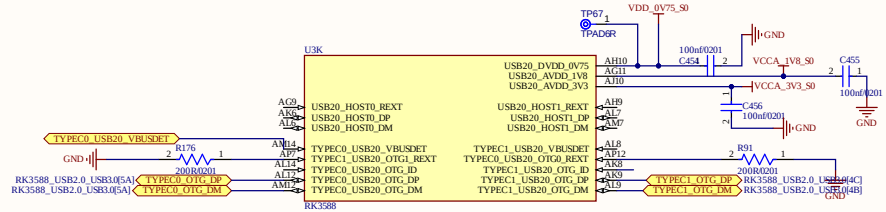
If not used: | If not used:
DP/DM:Leave floating | DP/DM:Leave floating
ID:Leave floating | REXT:Leave floating
VBUSDET:Leave floating
REXT:Leave floating

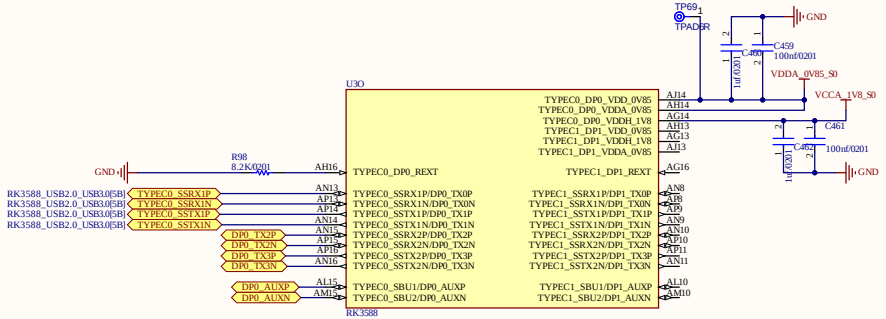
Note:

The USB20_VBUSDET pin internal has a pull-down resistance(400K ohm) to ground,The resistance creates a voltage with the external series 30K ohm resistor.The VBUSDET pin voltage range <=3.3V.

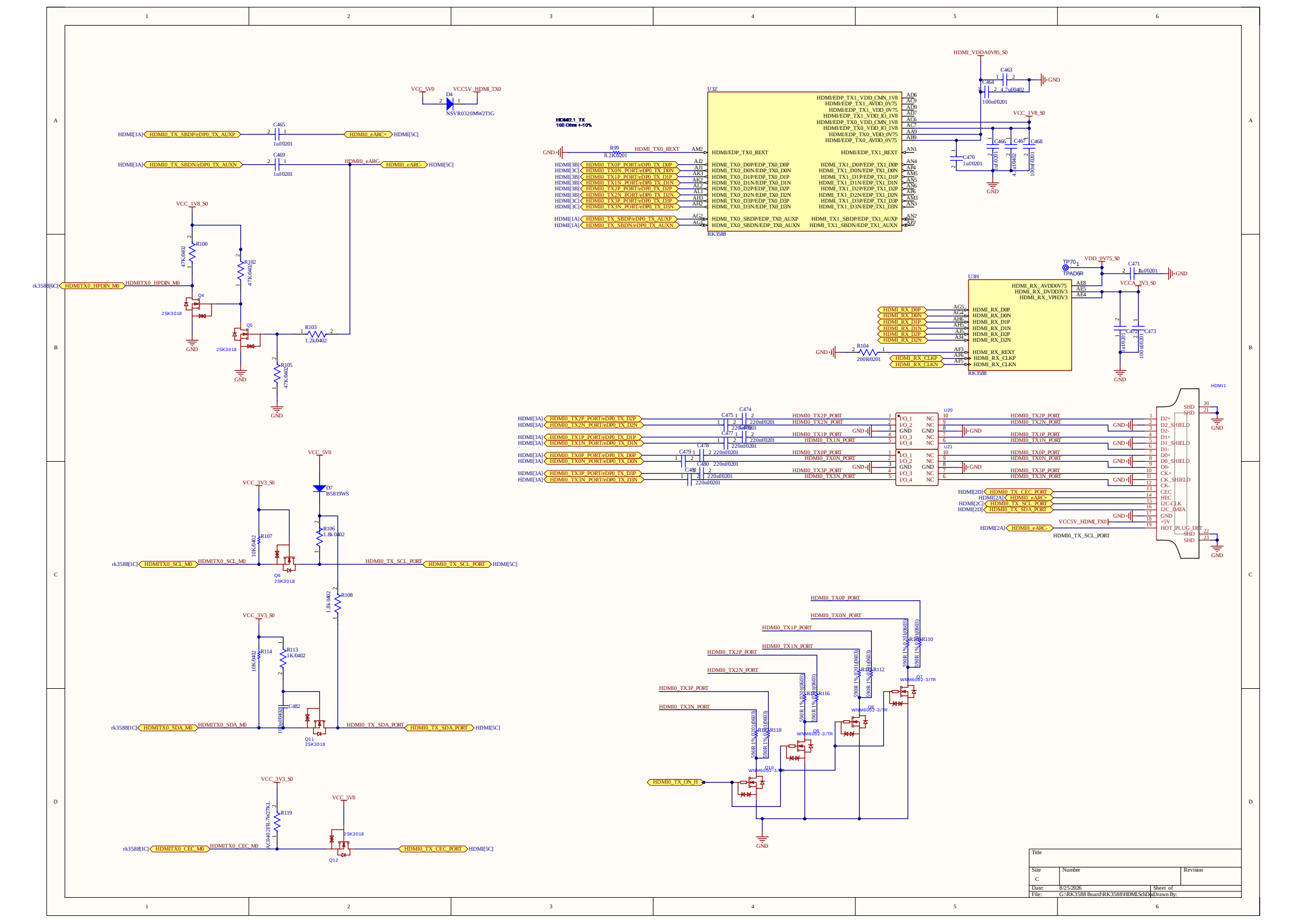


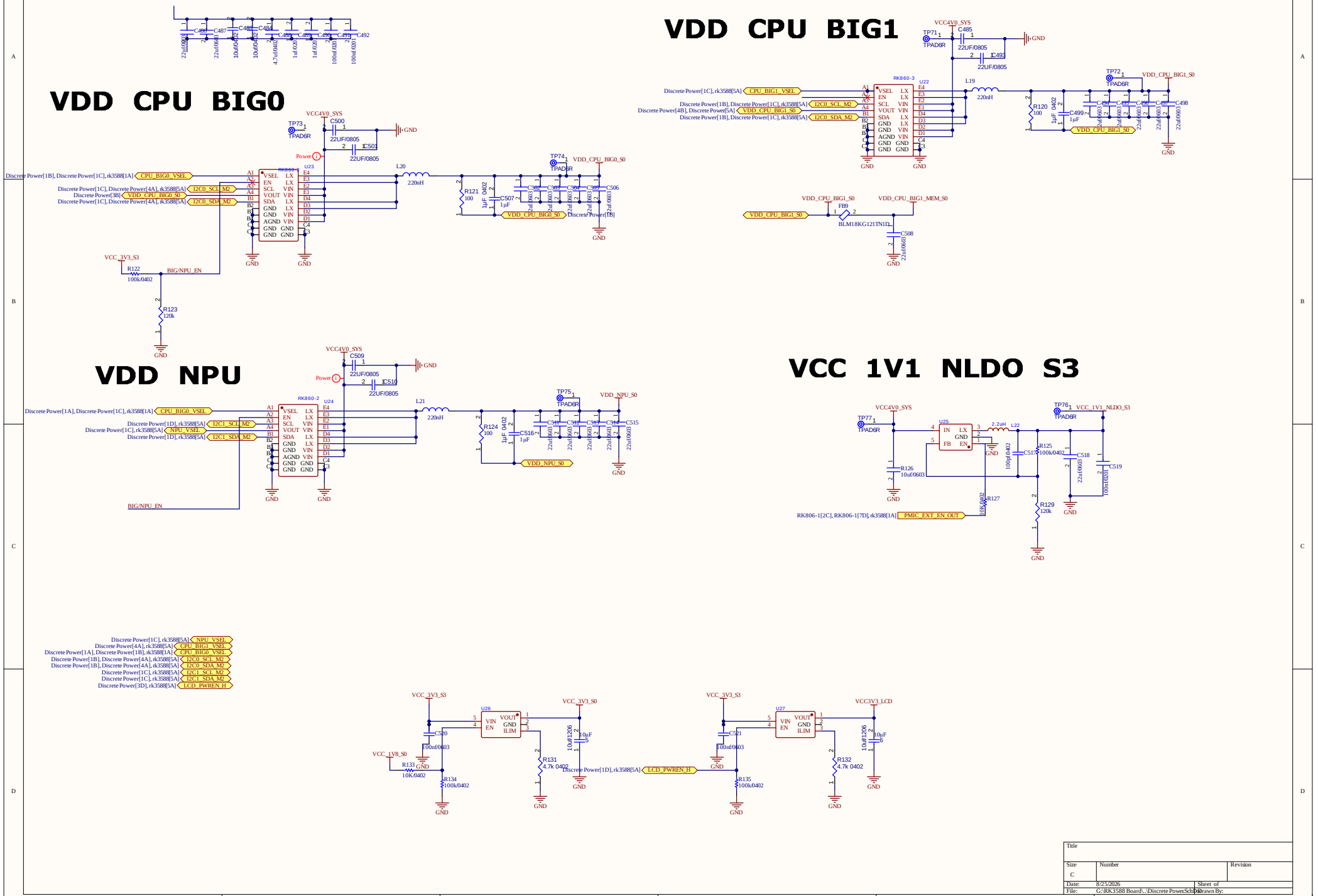


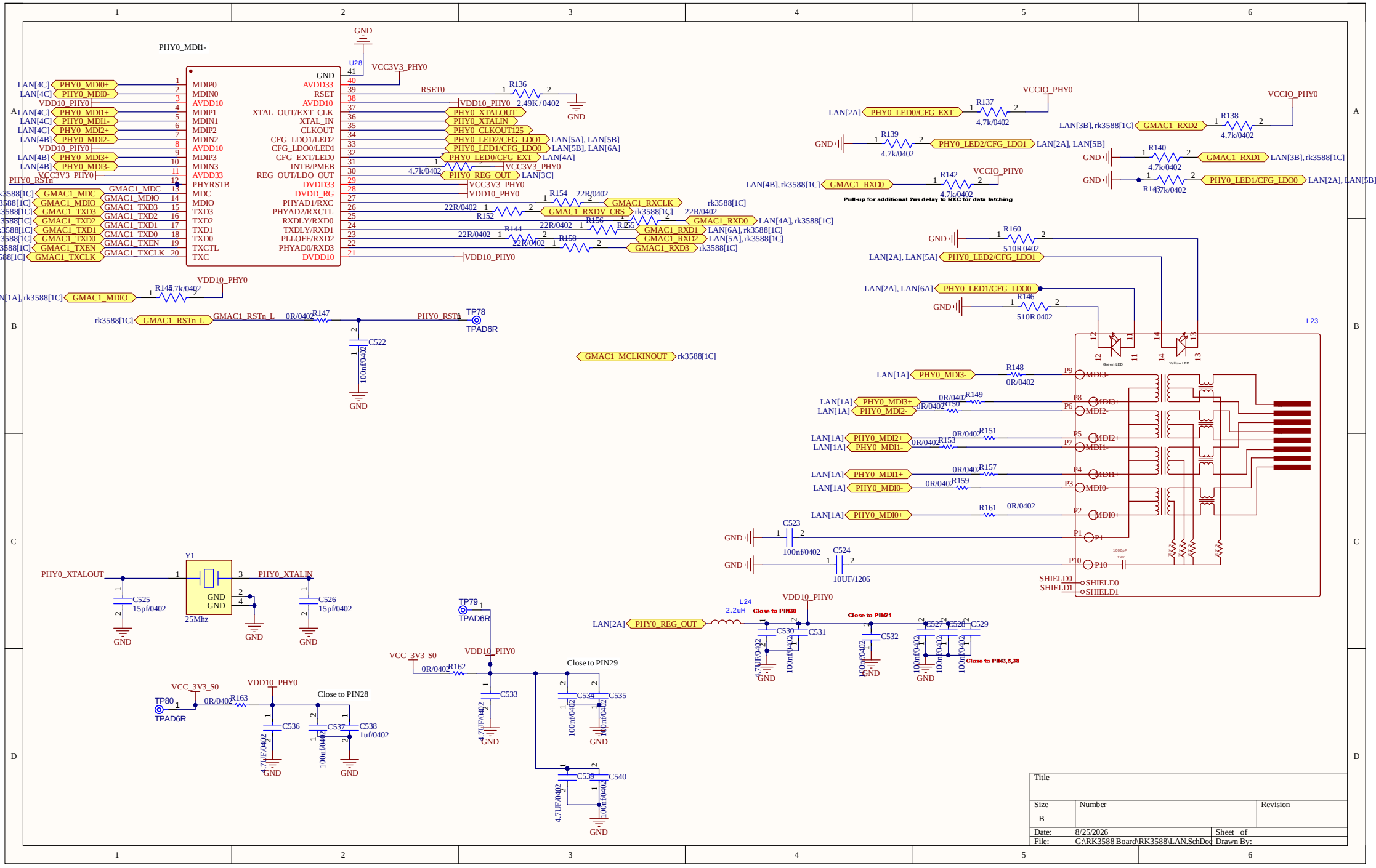


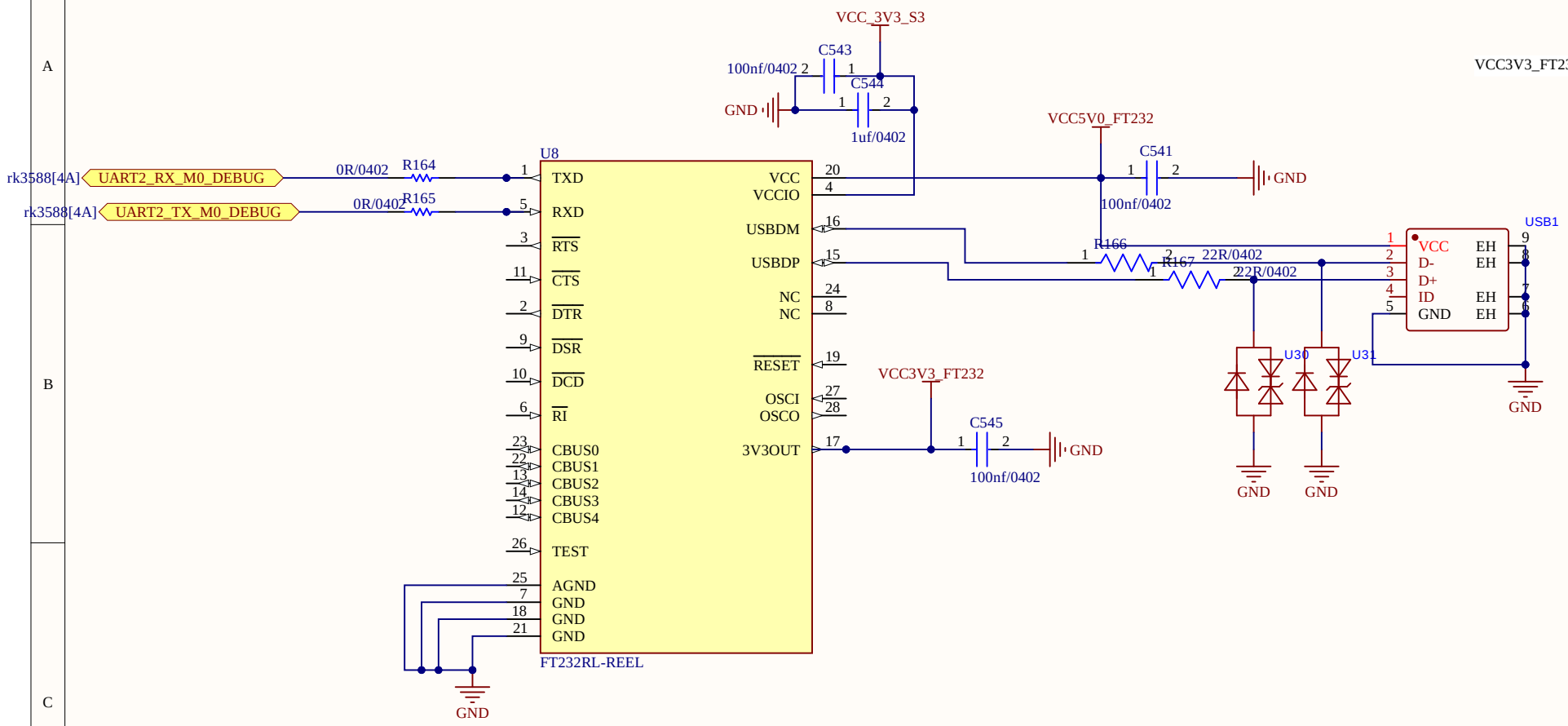


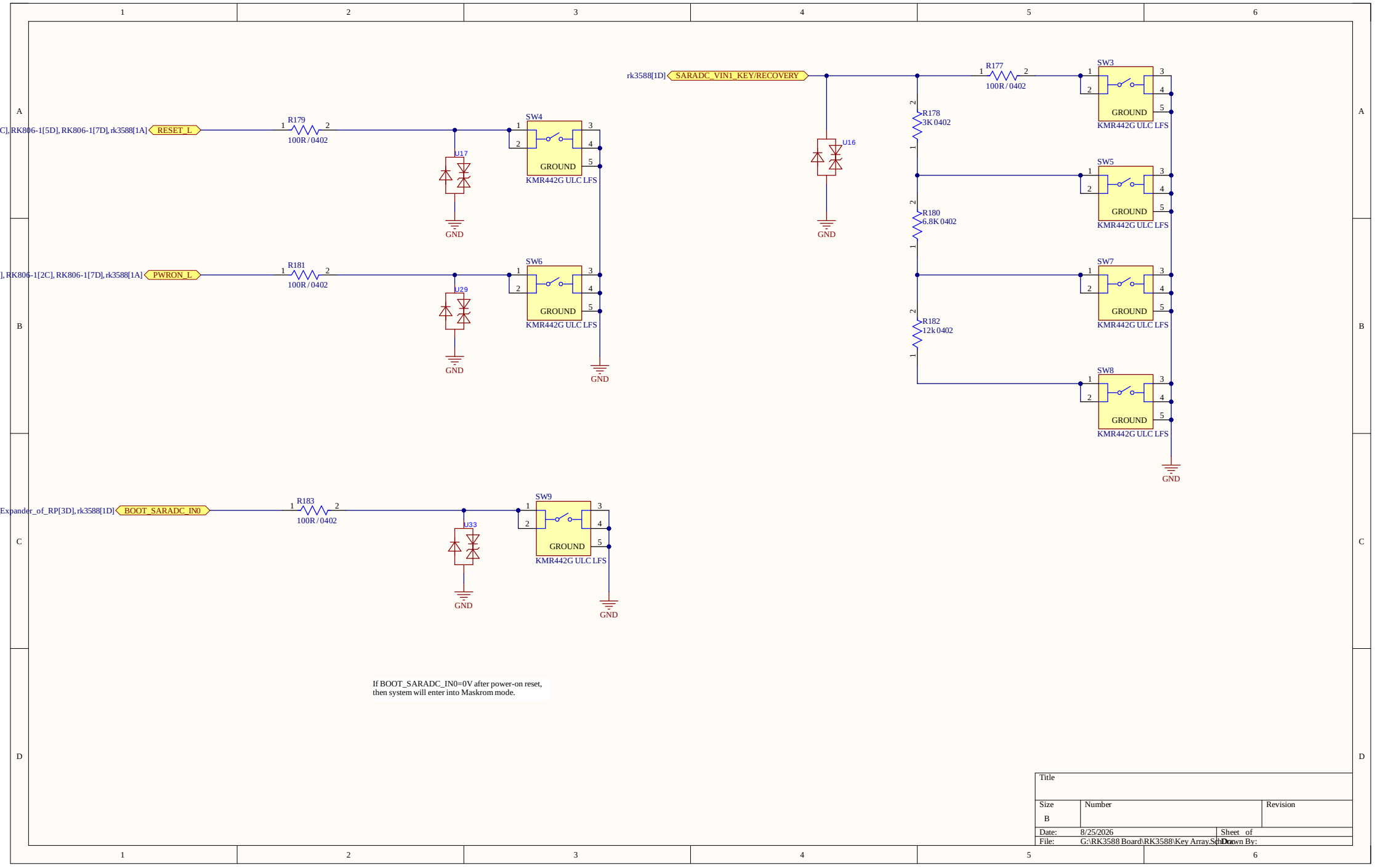
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Size: C	Number	Revision
Date: 8/25/2026	Sheet of	
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VCC_5V0
VCC5V_HDMI_TX0
D4
NSVR0320MW2T1G
HDMI[3A]
HDMI0_TX_SBDP/eDP0_TX_AUXP
C465
1uf/0201
HDMI0_eARC+
HDMI[5C]
HDMI[3A]
HDMI0_TX_SBDN/eDP0_TX_AUXN
C469
1uf/0201
HDMI0_eARC-
HDMI0_eARC-
HDMI[5C]
VCC_1V8_S0
R100
47K/0402
R102
47K/0402
rk3588[6C]
HDMITX0_HPDIN_M0
HDMITX0_HPDIN_M0
Q4
2SK3018
Q5
2SK3018
R103
1.2k/0402
R105
47K/0402
GND
GND
GND
HDMI_VDDA0V85_S0
C463
C464
4.7uf/0402
100nf/0201
GND
VCC_1V8_S0
C466
C467
C468
1uf/0201
4.7uf/0402
100nf/0201
C470
1uf/0201
GND
U3Z
HDMI/EDP_TX1_VDD_CMN_1V8
HDMI/EDP_TX1_AVDD_0V75
HDMI/EDP_TX1_VDD_0V75
HDMI/EDP_TX1_VDD_IO_1V8
HDMI/EDP_TX0_VDD_CMN_1V8
HDMI/EDP_TX0_VDD_IO_1V8
HDMI/EDP_TX0_VDD_0V75
HDMI/EDP_TX0_AVDD_0V75
HDMI/EDP_TX0_REXT
HDMI/EDP_TX1_REXT
HDMI_TX0_D0P/EDP_TX0_D0P
HDMI_TX0_D0N/EDP_TX0_D0N
HDMI_TX0_D1P/EDP_TX0_D1P
HDMI_TX0_D1N/EDP_TX0_D1N
HDMI_TX0_D2P/EDP_TX0_D2P
HDMI_TX0_D2N/EDP_TX0_D2N
HDMI_TX0_D3P/EDP_TX0_D3P
HDMI_TX0_D3N/EDP_TX0_D3N
HDMI_TX0_SBDP/EDP_TX0_AUXP
HDMI_TX0_SBDN/EDP_TX0_AUXN
HDMI_TX1_D0P/EDP_TX1_D0P
HDMI_TX1_D0N/EDP_TX1_D0N
HDMI_TX1_D1P/EDP_TX1_D1P
HDMI_TX1_D1N/EDP_TX1_D1N
HDMI_TX1_D2P/EDP_TX1_D2P
HDMI_TX1_D2N/EDP_TX1_D2N
HDMI_TX1_D3P/EDP_TX1_D3P
HDMI_TX1_D3N/EDP_TX1_D3N
HDMI_TX1_SBDP/EDP_TX1_AUXP
HDMI_TX1_SBDN/EDP_TX1_AUXN
RK3588
HDMI2.1_TX
100 Ohm ±10%
R99
8.2K/0201
GND
HDMI[3B] HDMI0_TX0P_PORT/eDP0_TX_D0P
HDMI[3C] HDMI0_TX0N_PORT/eDP0_TX_D0N
HDMI[3B] HDMI0_TX1P_PORT/eDP0_TX_D1P
HDMI[3B] HDMI0_TX1N_PORT/eDP0_TX_D1N
HDMI[3B] HDMI0_TX2P_PORT/eDP0_TX_D2P
HDMI[3B] HDMI0_TX2N_PORT/eDP0_TX_D2N
HDMI[3C] HDMI0_TX3P_PORT/eDP0_TX_D3P
HDMI[3C] HDMI0_TX3N_PORT/eDP0_TX_D3N
HDMI[1A] HDMI0_TX_SBDP/eDP0_TX_AUXP
HDMI[1A] HDMI0_TX_SBDN/eDP0_TX_AUXN
TP70
TPAD6R
VDD_0V75_S0
C471
1uf/0201
GND
U3H
HDMI_RX_AVDD0V75
HDMI_RX_DVDD3V3
HDMI_RX_VPH3V3
VCCA_3V3_S0
C472
C473
1uf/0201
100nf/0201
GND
HDMI_RX_D0P
HDMI_RX_D0N
HDMI_RX_D1P
HDMI_RX_D1N
HDMI_RX_D2P
HDMI_RX_D2N
HDMI_RX_REXT
HDMI_RX_CLKP
HDMI_RX_CLKN
RK3588
R104
200R/0201
GND
HDMI1
U20
U21
HDMI[3A] HDMI0_TX2P_PORT/eDP0_TX_D2P
HDMI[3A] HDMI0_TX2N_PORT/eDP0_TX_D2N
HDMI[3A] HDMI0_TX1P_PORT/eDP0_TX_D1P
HDMI[3A] HDMI0_TX1N_PORT/eDP0_TX_D1N
HDMI[3A] HDMI0_TX0P_PORT/eDP0_TX_D0P
HDMI[3A] HDMI0_TX0N_PORT/eDP0_TX_D0N
HDMI[3A] HDMI0_TX3P_PORT/eDP0_TX_D3P
HDMI[3A] HDMI0_TX3N_PORT/eDP0_TX_D3N
C474
C475
C476
C477
C478
C479
C480
C481
220nf/0201
HDMI0_TX2P_PORT
HDMI0_TX2N_PORT
HDMI0_TX1P_PORT
HDMI0_TX1N_PORT
HDMI0_TX0P_PORT
HDMI0_TX0N_PORT
HDMI0_TX3P_PORT
HDMI0_TX3N_PORT
I/O_1 NC
I/O_2 NC
GND GND
I/O_3 NC
I/O_4 NC
HDMI[2D] HDMI0_TX_CEC_PORT
HDMI[2A] HDMI0_eARC+
HDMI[2C] HDMI0_TX_SCL_PORT
HDMI[2D] HDMI0_TX_SDA_PORT
VCC5V_HDMI_TX0
HDMI[2A] HDMI0_eARC-
HDMI0_TX_SCL_PORT
D2+
D2_SHIELD
D2-
D1+
D1_SHIELD
D1-
D0+
D0_SHIELD
D0-
CK+
CK_SHIELD
CK-
CEC
HEC
I2C_CLK
I2C_DATA
GND
+5V
HOT_PLUG_DET
SHD
VCC_5V0
D7
B5819WS
VCC_3V3_S0
R106
1.8k/0402
R107
10K/0402
rk3588[1C]
HDMITX0_SCL_M0
HDMITX0_SCL_M0
HDMI0_TX_SCL_PORT
HDMI0_TX_SCL_PORT
HDMI[5C]
Q6
2SK3018
R108
1.8k/0402
VCC_3V3_S0
R113
1K/0402
R114
10K/0402
C482
10nf/0402
rk3588[1C]
HDMITX0_SDA_M0
HDMITX0_SDA_M0
HDMI0_TX_SDA_PORT
HDMI0_TX_SDA_PORT
HDMI[5C]
Q11
2SK3018
VCC_3V3_S0
R119
ACO402FR-7W27KL
VCC_5V0
2SK3018
Q12
rk3588[1C]
HDMITX0_CEC_M0
HDMITX0_CEC_M0
HDMI0_TX_CEC_PORT
HDMI[5C]
HDMI0_TX0P_PORT
HDMI0_TX0N_PORT
HDMI0_TX1P_PORT
HDMI0_TX1N_PORT
HDMI0_TX2P_PORT
HDMI0_TX2N_PORT
HDMI0_TX3P_PORT
HDMI0_TX3N_PORT
R110
R111
R112
R116
R117
R118
590R 1% 1/20W(0603)
Q7
Q8
Q9
Q10
WNM6002-3/TR
HDMI0_TX_ON_H
GND
Title
Size
C
Number
Revision
Date:
8/25/2026
Sheet of
File:
G:\RK3588 Board\RK3588\HDMI.SchDoc
Drawn By:


VDD CPU BIG0
VDD CPU BIG1
VDD NPU
VCC 1V1 NLDO S3
VCC4V0_SYS
VCC_3V3_S3
VCC_3V3_S0
VCC3V3_LCD
VCC_1V8_S0
VDD_CPU_BIG0_S0
VDD_CPU_BIG1_S0
VDD_NPU_S0
VCC_1V1_NLDO_S3
VDD_CPU_BIG1_MEM_S0
CPU_BIG0_VSEL
CPU_BIG1_VSEL
NPU_VSEL
I2C0_SCL_M2
I2C0_SDA_M2
I2C1_SCL_M2
I2C1_SDA_M2
LCD_PWREN_H
PMIC_EXT_EN_OUT
BIG/NPU_EN
RK860-3
RK860-2
GND
Title
Size
C
Number
Revision
Date: 8/25/2026
Sheet of
File: G:\RK3588 Board\..\Discrete Power.SchDoc
Drawn By:

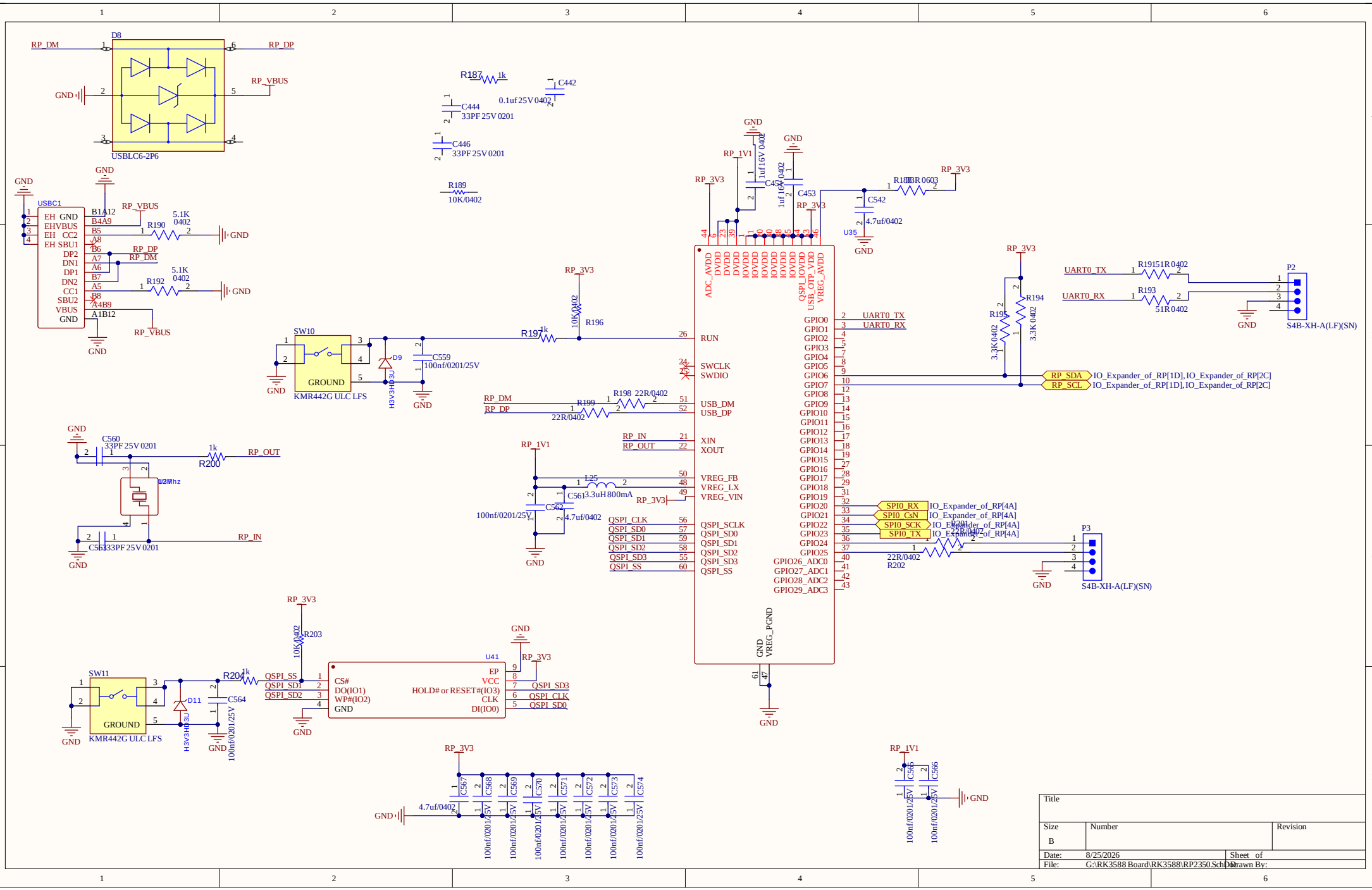

PHY0_MDI1-
U28
GND
VCC3V3_PHY0
MDIP0
MDIN0
AVDD10
MDIP1
MDIN1
MDIP2
MDIN2
AVDD10
MDIP3
MDIN3
AVDD33
PHYRSTB
MDC
MDIO
TXD3
TXD2
TXD1
TXD0
TXCTL
TXC
AVDD33
RSET
AVDD10
XTAL_OUT/EXT_CLK
XTAL_IN
CLKOUT
CFG_LDO1/LED2
CFG_LDO0/LED1
CFG_EXT/LED0
INTB/PMEB
REG_OUT/LDO_OUT
DVDD33
DVDD_RG
PHYAD1/RXC
PHYAD2/RXCTL
RXDLY/RXD0
TXDLY/RXD1
PLLOFF/RXD2
PHYAD0/RXD3
DVDD10
RSET0
R136
2.49K/0402
VDD10_PHY0
PHY0_XTALOUT
PHY0_XTALIN
PHY0_CLKOUT125
PHY0_LED2/CFG_LDO1
PHY0_LED1/CFG_LDO0
PHY0_LED0/CFG_EXT
PHY0_REG_OUT
VCC3V3_PHY0
VDD10_PHY0
4.7k/0402
LAN[5A], LAN[5B]
LAN[5B], LAN[6A]
LAN[4A]
LAN[3C]
PHY0_RSTn
GMAC1_MDC
GMAC1_MDIO
GMAC1_TXD3
GMAC1_TXD2
GMAC1_TXD1
GMAC1_TXD0
GMAC1_TXEN
GMAC1_TXCLK
R154 22R/0402
R152 22R/0402
R144 22R/0402
R158 22R/0402
R156 22R/0402
R155 22R/0402
GMAC1_RXCLK
GMAC1_RXDV_CRS
GMAC1_RXD0
GMAC1_RXD1
GMAC1_RXD2
GMAC1_RXD3
rk3588[1C]
LAN[4A], rk3588[1C]
LAN[6A], rk3588[1C]
LAN[5A], rk3588[1C]
R145 4.7k/0402
GMAC1_RSTn_L
R147 0R/0402
PHY0_RSTn
TP78
TPAD6R
C522
100nf/0402
GMAC1_MCLKINOUT
VCCIO_PHY0
R137 4.7k/0402
R138 4.7k/0402
R139 4.7k/0402
R140 4.7k/0402
R142 4.7k/0402
R143 4.7k/0402
LAN[2A]
LAN[3B], rk3588[1C]
LAN[2A], LAN[5B]
LAN[3B], rk3588[1C]
LAN[2A], LAN[5B]
LAN[4B], rk3588[1C]
Pull-up for additional 2ns delay to RXC for data latching
R160 510R 0402
R146 510R 0402
LAN[2A], LAN[5A]
LAN[2A], LAN[6A]
L23
Green LED
Yellow LED
R148 0R/0402
R149 0R/0402
R150 0R/0402
R151 0R/0402
R153 0R/0402
R157 0R/0402
R159 0R/0402
R161 0R/0402
LAN[1A]
PHY0_MDI3-
PHY0_MDI3+
PHY0_MDI2-
PHY0_MDI2+
PHY0_MDI1-
PHY0_MDI1+
PHY0_MDI0-
PHY0_MDI0+
MDI3-
MDI3+
MDI2-
MDI2+
MDI1-
MDI1+
MDI0-
MDI0+
P1
P2
P3
P4
P5
P6
P7
P8
P9
P10
SHIELD0
SHIELD1
C523 100nf/0402
C524 10UF/1206
1000pF 2KV
Y1
25Mhz
PHY0_XTALOUT
PHY0_XTALIN
C525 15pf/0402
C526 15pf/0402
TP79
TPAD6R
VCC_3V3_S0
R162 0R/0402
C533
C534
C535
Close to PIN29
TP80
TPAD6R
R163 0R/0402
C536
C537
C538 1uf/0402
Close to PIN28
L24 2.2uH
Close to PIN30
Close to PIN21
Close to PIN3,8,38
C530
C531
C532
C527
C528
C529
C539
C540
4.7UF/0402
100nf/0402
Title
Size
B
Number
Revision
Date: 8/25/2026
Sheet of
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Drawn By:

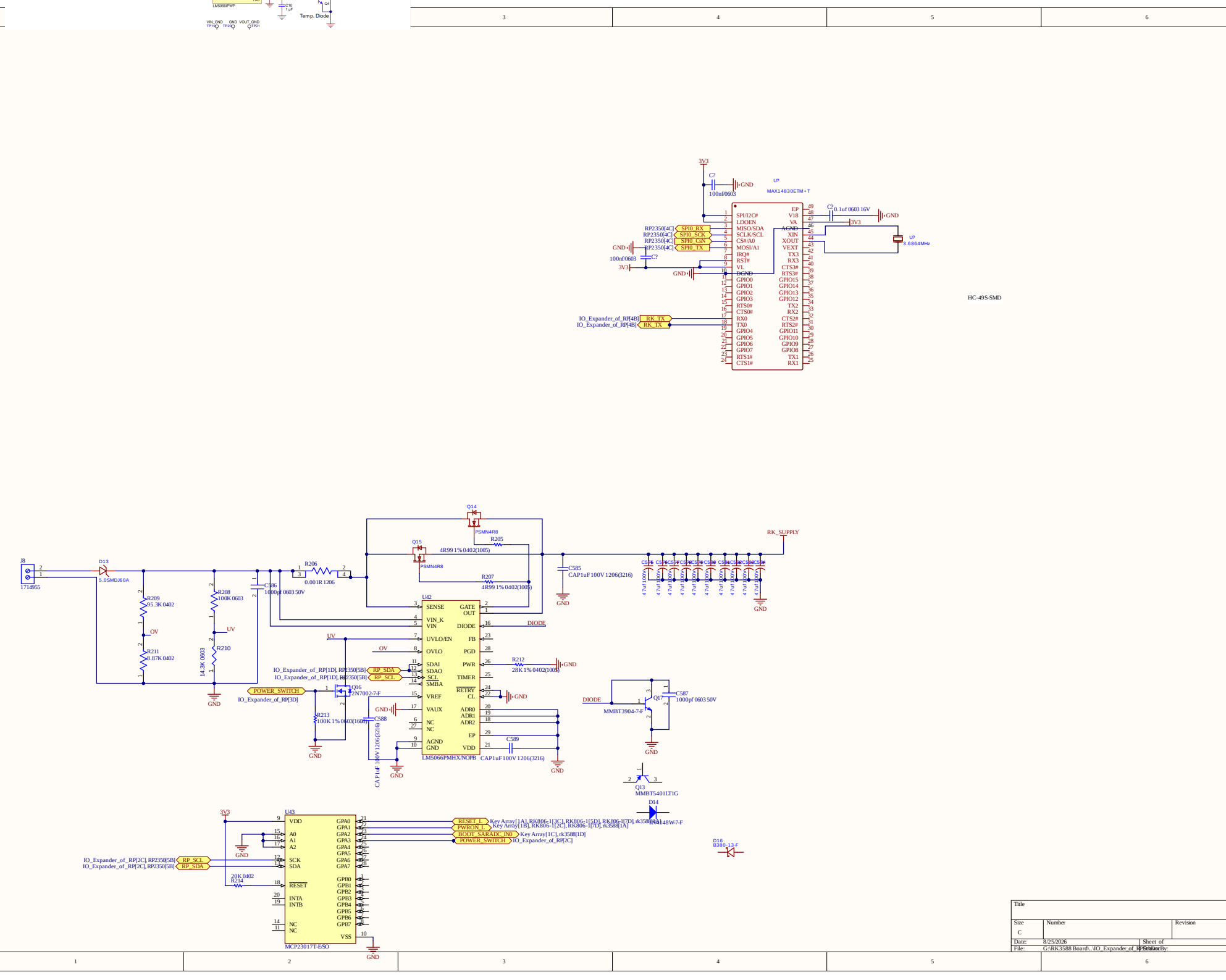

VCC_3V3_S3
C543
100nf/0402
C544
1uf/0402
GND
VCC5V0_FT232
C541
100nf/0402
VCC3V3_FT232
C545
100nf/0402
U8
FT232RL-REEL
TXD
RXD
RTS
CTS
DTR
DSR
DCD
RI
CBUS0
CBUS1
CBUS2
CBUS3
CBUS4
TEST
AGND
VCC
VCCIO
USBDM
USBDP
NC
NC
RESET
OSCI
OSCO
3V3OUT
rk3588[4A]
UART2_RX_M0_DEBUG
UART2_TX_M0_DEBUG
R164
R165
0R/0402
R166
R167
22R/0402
USB1
U30
U31
VCC
D-
D+
ID
GND
EH
VCC3V3_FT232
SIT3232EEUE
U32
C548
C546
C549
C550
C547
100nf/0402
C1+
V+
C1-
C2+
C2-
V-
T2OUT
R2IN
VCC
GND
T1OUT
R1IN
R1OUT
T1IN
T2IN
R2OUT
R168
R169
R170
R171
UART3_RX_M2
UART3_TX_M2
rk3588[1C]
P1
S4B-XH-A(LF)(SN)
VCC_3V3_S0
VCC3V3_FT232
Title
Size
A
Number
Revision
Date:
8/25/2026
Sheet of
File:
G:\RK3588 Board\..\RS232 DBG_UART_SCH.Dsn
Drawn By:



If BOOT_SARADC_IN0=0V after power-on reset,
then system will enter into Maskrom mode.

Title		
Size: B	Number:	Revision:
Date: 8/25/2026	Sheet of	
File: G:\RK3588 Board\RK3588\Key Array.SchDoc	Drawn By:	


D8
USBLC6-2P6
RP_DM
RP_DP
RP_VBUS
GND
USBC1
EH_GND
EHVBUS
EH_CC2
EH_SBU1
DP2
DN1
DP1
DN2
CC1
SBU2
VBUS
GND
B1A12
B4A9
B5
A8
B6
A7
A6
B7
A5
B8
A4B9
A1B12
R190 5.1K 0402
R192 5.1K 0402
R187 1k
C442 0.1uf 25V 0402
C444 33PF 25V 0201
C446 33PF 25V 0201
R189 10K/0402
SW10
GROUND
KMR442G ULC LFS
D9
H3V3H03U
C559 100nf/0201/25V
R196 10K/0402
R197 1k
RP_3V3
RP_1V1
1uf 16V 0402
C455
C453
C542 4.7uf/0402
R188 3R 0603
U35
ADC_AVDD
DVDD
IOVDD
QSPI_IOVDD
USB_OTP_VDD
VREG_AVDD
RUN
SWCLK
SWDIO
USB_DM
USB_DP
XIN
XOUT
VREG_FB
VREG_LX
VREG_VIN
QSPI_SCLK
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SS
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
GPIO24
GPIO25
GPIO26_ADC0
GPIO27_ADC1
GPIO28_ADC2
GPIO29_ADC3
VREG_PGND
UART0_TX
UART0_RX
R194
R195 3.3K 0402
R193 51R 0402
R191 51R 0402
P2
S4B-XH-A(LF)(SN)
RP_SDA
RP_SCL
SPI0_RX
SPI0_CsN
SPI0_SCK
SPI0_TX
IO_Expander_of_RP[1D], IO_Expander_of_RP[2C]
IO_Expander_of_RP[4A]
R201 22R/0402
R202 22R/0402
P3
C560 33PF 25V 0201
R200 1k
RP_OUT
RP_IN
C561 3.3uH 800mA
C562 4.7uf/0402
100nf/0201/25V
R198 22R/0402
R199 22R/0402
SW11
R204 1k
D11
C564
R203 10K/0402
U41
CS#
DO(IO1)
WP#(IO2)
HOLD# or RESET#(IO3)
CLK
DI(IO0)
EP
VCC
C567
C568
C569
C570
C571
C572
C573
C574
4.7uf/0402
C565
C566
Title
Size B
Number
Revision
Date: 8/25/2026
Sheet of
File: G:\RK3588 Board\RK3588\RP2350.SchDoc
Drawn By:



Title		
Size: C	Number	Revision
Date: 8/25/2026	Sheet of	
File: G:\RK3588 Board\..\IO_Expander_of_RP.SchDoc	Drawn By:	